

Chapter 1

Calibration of a UV/VIS Spectrometer

1.1 Introduction

The goal of this experiment is to calibrate an ultraviolet/visible (UV/VIS) spectrometer for use in determining concentrations of a compound in a solvent. The objectives are to learn how to operate a UV/VIS spectrometer, produce a calibration curve of amount of light absorbed by a solution versus solution concentration, confirm the applicability of the Beer-Lambert law to the data in calibration curve, and use the calibration curve to find the concentration of an unknown solution.

In UV/VIS spectroscopy, light in the wavelength range of ultra-violet is sent through a sample such as a liquid solution. Compounds in the sample absorb certain wavelengths of the light. The amount (intensity) of light absorbed is related to the concentration of the component in the sample. At sufficiently low concentrations of the absorbing component in the sample, a linear relationship exists between absorbance and concentration of material. This is expressed by the Beer-Lambert law

$$A = -\log(I/I_o)$$

$$A = \epsilon L [C]$$

where A is the absorbance, I is the intensity of transmitted light, I_o is the incident intensity, ϵ is the extinction coefficient of the material (compound in the sample), L is the path length through the sample (m), and $[C]$ is the concentration of the species under consideration (mol/L).

A typical UV/VIS spectrum is a plot of the amount of light absorbed by a sample (I) versus the wavelength of light. This can be converted to absorbance A using the initial intensity of the light I_o . A peak will appear in an absorbance spectrum at a wavelength where the light is absorbed by the compound. In a rigorous treatment of the subject, the integrated area under this peak is related to the concentration of the compound in the sample. As a good first approximation, the height of the peak at a specific (constant) wavelength can be used in place of the area under the peak.

Liquid samples are typically put into a solution cell called a cuvette for measurement of the absorbance. The standard configuration of a cuvette for a UV/VIS spectrometer is shown schematically on the right. In cases where the path length is always constant (as with this experiment), the value of ϵ and L can be combined to a term α called the absorption coefficient.

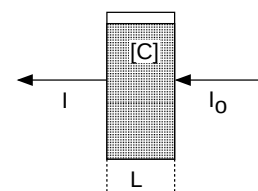


Figure 1.1: Geometry for a UV/VIS measurement.

Although the extinction coefficient of a compound ϵ is a property of the compound and its solvent, the absorption coefficient will vary from spectrometer to spectrometer (why?). Therefore, a UV/VIS spectrometer has to be calibrated to obtain the adsorption coefficient for a given compound of interest.

Calibration is the process of obtaining the response of the system (the spectrometer in this case) to a known input (a variation in solution concentration). A calibration curve is a plot of the response versus the input. A calibration curve for a compound measured with UV/VIS spectroscopy

would be a plot of absorbance versus solution concentration. Such a curve is typically then used to determine the concentration in an unknown solution by measuring its absorbance A .

1.2 Experimental Methods

You will be given a solution with a compound in a solvent, typically a dye in water. You will prepare six solutions at different concentrations of the compound in solvent using standard dilution methods. You will subsequently obtain the UV/VIS spectra of the solutions. The absorbance at a certain wavelength will be plotted versus solution concentration. From the plot, you will determine the absorption coefficient of the solution. Finally, you will be given a solution with an unspecified concentration of the compound in solvent and asked to determine the concentration of the solution.

Supplies

- a flask containing a known concentration C_o of dye in a solvent
- twelve 150 ml test tubes
- a test tube rack
- two 10 ml pipettes
- two 1 ml pipettes
- two 50 ml beakers
- two 100 ml beakers
- UV/VIS spectrophotometer
- cuvettes for the spectrophotometer

Procedures

Starting from the stock solution of known concentration C_o , prepare a larger volume of standard solution at $0.1C_o$. Determine the dilution steps needed to prepare five other solutions at decreasing concentrations. A dilution step consists of taking a given amount of a solution with a known concentration and adding a given amount of solvent to decrease the concentration to a desired value. Unless instructed otherwise, you are to prepare solutions at concentrations of $0.1C_o$ (the standard), $0.02C_o$, $0.01C_o$, $0.002C_o$, $0.001C_o$, and $0.0002C_o$. This spans four orders of magnitude in solution concentration.

Two methods of dilution are to be used. In the first, you are to prepare all solutions from the standard solution at $0.1C_o$. In the second, you are to prepare each solution from the solution at the next higher concentration. The dilution equation for either process is

$$C_f = \frac{C_i V_s}{V_f}$$

where C_f is the desired final concentration, C_i is the initial concentration of the solution, V_s is the volume of the initial solution, and V_f is the volume of the final solution. The final volume of solution is the initial V_i plus the amount of solvent added.

You are to record the UV/VIS spectra for the six solutions prepared by each method. This will give a total of twelve spectra. The lab instructor will discuss the details of how to record the UV/VIS spectra using the instruments available in the lab. To start recording your data, choose a clean cuvette. Record and store the spectrum of the pure solvent. This is called a blank. The blank will

have to be removed from all subsequent spectra of the solutions. The instrument can generally remove the blank from the subsequent sample spectra automatically. The blank is also used to obtain the reading for I_0 , the initial intensity of light.

After recording the blank, continue with the lowest concentration of solution. Use the same cuvette throughout (why?). Rinse out the cuvette with a small amount the solution and discard the solution. Repeat this process two more times. This prepares the cuvette for the solution (it assures that none of the solvent from the previous run remains). Measure the absorbance spectrum for the solution. Store the spectrum of the solution. Repeat the above measurements for each of the remaining solutions in the series.

Before starting on the second series of concentrations, re-record the blank.

Reproducibility and Accuracy

You are required to repeat the solution preparation and measurement steps at least three times for each of the two dilution sequences.

This experiment does not require confirmation of its accuracy.

Cautions

Clean and use the same cuvette throughout the entire experiment. Do not discard it. Also, try to use the same UV/VIS spectrometer throughout the experiment. One method to check the reproducibility of the UV/VIS spectrometers from one week to the next is to use a stock solution and confirm that it gives the same absorbance every time. Finally, keep and use the initial concentrated solution that you are given at the start throughout the entire experiment.

1.3 Reports

First Week Report

Results

Figures

Show one plot of a typical UV/VIS spectrum as absorbance versus wavelength.

Plot absorbance versus concentration for all runs you do. Plot each dilution series on its own graph.

On each plot, fit the data with a straight line. Determine the slope and its uncertainty.

Tables

Tabulate the absorption constant α for each of the runs.

Other

Report the average absorption coefficient.

Report the standard error of the absorption coefficient.

Using the uncertainties of the slopes as measurement uncertainties, report an average measurement uncertainty for the absorption coefficient.

Report a final value of absorption coefficient with one standard error uncertainty.

Final Report

Results

Tables

For each run that has been done, tabulate the absorption constant α and its measurement uncertainty.

For each of the two dilution methods used, tabulate the average absorption coefficient from all runs with one standard error uncertainty, incorporating random and measurement uncertainties.

Figures

Show a typical UV/VIS spectrum.

For each of the two dilution methods used, plot average absorbance versus concentration. Include error bars on each data point to indicate the uncertainty in absorbance obtained for the given concentration.

Plot absorbance versus concentration for all runs you do individually, each on its own graph.

Plot absorbance versus concentration for the average of the runs.

Other

Report the following values: the average Beers Law constant with its standard deviation.

Sample Calculations

Show one example of the calculation used to obtain molar solution concentration from information about the mass of dye per volume of solution.

Sample Uncertainty Analysis

Show the calculation of the uncertainty on the average value of the Beers Law constant.

1.4 Summary Questions

Be prepared to answer any of these questions at the start of the lab session. You may need to read additional references to answer these questions.

1. What is the Beer-Lambert law?
2. What are the assumptions behind the Beer-Lambert law and how do they limit its applicability in measuring or determining concentration?
3. What causes a compound in a liquid solvent to absorb UV light?
4. What is a blank? When is it recorded? Why is it recorded?
5. What is the typical energy range of light for UV/VIS spectroscopy?